

Chapter Thirteen

Food and Nutrition

At present there are thousands of species of animals living in this world. They show variation in their shape-size and characteristics as well as in life style, nature, food and feeding habits. Food is very essential for growth, energy and livelihood of every animal. Therefore, food is also essential for sound and healthy human body. Adequate knowledge about food and nutrition is the pre-condition of sound and healthy body. We eat organic compounds like proteins, carbohydrates, fats and oils etc. as food. We get nutrition from these food. Organic compounds which are used by organisms for their growth, repair and production of energy are called food. From these foods, living beings get obtain the nutrition they need.



At the end of this chapter, we will be able to -

- explain the nutritional value of different foods;
- describe different means to prevent malnutrition-related diseases;
- select food items based on requirements.

Lesson 1: Nutrition, Nutrition Value and Food Elements

To run an engine, fuels like coal, diesel, petrol, natural gas etc. are used. Can you say what these fuels are used for? These fuels are burnt to produce energy. This energy provides power to the vehicles. The vehicles are kept running by this energy. The human body is compared to an engine. Like other engines, energy is needed to run our body. Where do we get these energy from? Food fulfils the need for nutrition in our body and supplies energy.

The living organism is the main source of food. Food is mainly composed of different compounds. We get food mainly from plants and animals. By food we mean the organic elements used for the growth of body and production of energy for living beings. The ingredients of food or nutritive substances present in food perform mainly three functions of our body. For example

- Growth, repair and maintenance of the body.
- Providing heat and calorie.
- Preventing disease, ensuring sound health and controlling metabolic activities (such as digestion, respiration, excretion etc).

Nutrition and Nutrition Value

Nutrition is a process. In this process, food is digested after eating and complex food ingredients are broken down into simple ingredients suitable for absorption. The body absorbs these simple ingredients. After absorption, the nutrients reach all the cells of the body. As a result, every part of the body regenerates the damaged cells and forms new cells for the growth of the body. Besides, food also produces heat, provides nutrition to prevent diseases and maintenance of our body. All these activities of the body are part of the nutrition process. Therefore, the nutritional elements are the valuable components of foods. That provide energy and appropriate growth of the body, enhance merit and intelligence, prevent diseases, help for quick recovery from illness and make human beings active.

The nutritional value of any food depends on the types and amounts of elements present in it. For example, boiled rice contains 79% carbohydrates and 6% fats. Additionally, it contains a minute amount of proteins, vitamins and mineral salts. A 100-gram serving of rice provides about 345-349 kilocalories. Boiled rice contains carbohydrates, proteins, and vitamins, but it has a higher carbohydrate content. Therefore, rice is mainly a carbohydrate-based food.

In order to know the nutritional value of any food we need to know the nature of the food. The nature of food means whether it is a mixed food or pure one. In a mixed food, more than one nutrition elements are present. For example, egg, milk, hotchpotch, guava, etc. On the other hand, in a pure food only one element is found. For example, sugar, glucose. There are no other elements except carbohydrate in it.

Food Elements

Food is composed of a number of chemical elements. These chemical elements are called food nutrients. There is hardly any food which is made up of only one element. Based on its nutrients food is classified into three main groups. For example-

1. Proteins – repair, growth and developments of the body.
2. Carbohydrates – produce energy.
3. Lipids or fats – generates heat and energy.

Besides these, three other nutrients are also essential. For example-

1. Vitamins – helps in prevention of disease, increases energy and stimulates different physio-chemical reactions.
2. Mineral salts – they take part in different physiological activities.
3. Water – maintains water balance of the body, controls the qualities of the cells, and helps in thermoregulation.

Lesson 2-3 : Carbohydrate and Protein

Carbohydrate

We usually eat ruti, puffed rice (muri), chira, bread for breakfast. These are sources of carbohydrates. Carbohydrates are energy producing food. Among the different elements of our daily food, carbohydrate dominates. The carbohydrates are easily digested. All carbohydrates are composed of three basic elements, carbon, hydrogen and oxygen. In our body carbohydrates provide energy for work. Glucose is a kind of simple carbohydrate. Based on chemical composition, carbohydrates are divided into three groups. Only one carbohydrate molecule makes monosaccharide. This is also called simple carbohydrate. During digestion, disaccharides and polysaccharide are broken down into simple carbohydrates (monosaccharides) and then absorbed into the body. For proper nutrition of the body, simple carbohydrates are very important because human body, can only absorb them. Of the three main carbohydrates - glucose, fructose and galactose; glucose gets into the whole body through blood stream.

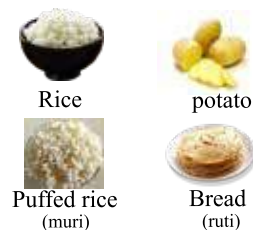


Fig. 13.1: Carbohydrate foods

Among carbohydrates, lipids and proteins, carbohydrate are the easiest to digest. Once absorbed in the body, they generate heat very quickly and supply energy to the body. One gram of carbohydrate can produce 4 kilocalories of heat. In a human body 300-400 grams of carbohydrates can be stored. This quantity of carbohydrate can produce 1200-1600 kilocalories heat and supply energy to our body.

The demand for carbohydrates depend on age, weight of the body, height and the amount of physical activity. An adult male person needs 4.6 grams of carbohydrates per kilogram of his body weight. An adult person weighing 60 kg requires (60×4.6) grams or 276 grams of carbohydrates daily on an average. Of the total requirement, we should intake 60-70 percent of calories from carbohydrate.

Activity : Identification of carbohydrate.

Necessary components : Rice starch, test tube, iodine, water, dropper etc

Methodology: Take a small amount of or rice starch into a test tube and mix little water in it. Now pour two to three drops of iodine solution into it. Observe what happens. The solution turns blue. From this experiment, the presence of carbohydrates can be determined.

Diseases Caused by Deficiencies

The intake of carbohydrate, be it more or less, both have negative impact on our body. Due to deficiency of carbohydrate malnutrition occurs. Fall of carbohydrate in blood creates metabolic disorder in the body. If there is a fall of carbohydrate in blood, symptom of hypoglycemia appears. For example—Feeling of appetite, Nausea, Excessive sweating, and Irregular heart beat.

Proteins

Protein is the composite element of our body. It is composed of carbon, hydrogen, oxygen and nitrogen. Proteins contain 16% of nitrogen. Therefore, in nutrition science protein is considered as a very important food nutrient. Protein is a complex compound of amino acids. During digestion, it turned into amino acids, suitable for the body to absorb. As of now a total of 22 types of amino acids have been found in the nature. The way we arrange Bangla or English alphabets to get numerous words, the proteins are produced in the same way after arranging 22 amino acids in different ways, number and types. For this reason, there are differences in the taste, smell, and colour of foods such as fish, milk, meat etc.

Some amino acids are very important for the growth, repair and balance of nitrogen of the body. These are called essential amino acids. These amino acids cannot be produced in the body. They must be obtained from food.

In food, if there is a deficiency of animal proteins i.e. essential amino acids, symptoms of various diseases arises. For example—nausea, increase of organic acid in urine, imbalance of nitrogen, etc.



Fig. 13.2: Protein foods

All proteins are not equally absorbed in the body. The digestibility factor of a protein is the percentage absorbed from the intestines after consumption. The nutritional value of proteins depends on their digestibility. After consuming proteins, the percentage of protein absorbed from the intestine is called the easily digestible quotient of the proteins. Proteins that are 100 percent absorbed by the body and promote growth and repair have an easily digestible quotient of 1. In this case, protein intake and absorption are equal. Simply put, the amount of protein taken by the organism is fully used by the body for growth and repair. If this is not the case, the easily digestible quotient is less than 1. The easily digestible quotient of breast milk and eggs is 1. The easily digestible quotient of other proteins is less than 1.

Activity : Identification of protein.

Necessary components : Egg albumin, spoon, water, test tube, sodium hydroxide, copper sulphate etc

Methodology : A little amount of protein (egg albumin) has to be stirred with the help of a spoon. For smooth stirring a little amount of water may be added to it. Now take a small amount of protein solution in the test tube. In this solution, add few drops of sodium hydroxide and copper sulphate solution. Can you see any changes in the solution?

The solution has taken violet colour following the mixing of the chemical elements with protein solution. This indicates the presence of protein.

Diseases Caused by Protein Deficiency

The body of children suffers from different protein deficiency related complications if food lacks sufficient proteins. The usual growth and developments of the body are also hampered. When children suffer from malnutrition, the growth of the body is stopped or delayed. The children suffer from kwashiorkor or marasmus disease due to protein deficiency.

Symptoms of Kwashiorkor Diseases

- Children lose their appetite.
- The muscle becomes scaly and weak, Moreover, the skin and hair lose their smoothness and colour.
- Children suffer from diarrhoea, and oedema.
- Stomach swells up.

Through proper treatment this disease can be cured, yet a kind of mental depression persists. If kwashiorkor becomes acute, it may cause death of the child.

Symptoms of Marasmus Disease

- The growth of the body is retarded due to deficiency of both proteins and calories.
- The body becomes lean and thin and reduced to a skeleton.
- The skin becomes rough and hanging down.
- Body loses the weight.

This state is dangerous for the children. Due to protein deficiency the elderly people lose their body's resistance and develop anemia.

Lesson 4-5 : Lipids

This is called energy producing ingredients. In lipids carbon, hydrogen and oxygen are found more in quantity. Since carbon has higher combustion quality the molecules of lipid substance release more heat energy. Lipids are a compound made up of fatty acid and glycerol. The fatty substance having digested turn into fatty acid and glycerol.

The fatty acids and glycerol are absorbed in the body through the lymph glands located in the villi of the intestine. In lipids 20 types of fatty acids are found. Fatty acids are of two types. For example: 1. Unsaturated fatty acids and 2. Saturated fatty acids.

Fatty acids are formed in liver. But the capacity of liver to produce fatty acid is very limited. On the other hand, there are some fatty acids essential for the body. These are mainly of plant origin. The usefulness of it cannot be measured by the quantity of fats present in food. Food, that contains more unsaturated fatty acids are more useful. For example: Soyabean oil, sunflower oil, corn oil, sesame oil etc. The food made with these oils fall under the group of best fatty food. For example: mayonnaise, salad dressing, kasundi, oil pickles are included in the group of fatty food. The foods that contain more saturated fatty acids are called fat enriched food. For example- meat, cheese, butter, dalda, chocolate, nuts, etc. According to nutrition science out of the daily required energy for the body, 20% - 30% energy is obtained from fats. We should include those fatty food in our daily food menu which can supply essential fatty acids and can dissolve vitamins in solution.

If food lacks fatty substances, deficiency of fat soluble vitamins occurs in the body. This will result in diseases due to the deficiency of vitamin. For example: skin will become dry and rough that will ruin the beauty of the body, the children will suffer from eczema caused due to the deficiency of essential fatty acids, and elderly people lose resistance against skin disease.



Fig. 13.3: Fatty foods

Activity : Identification of lipids.

Necessary components : Soyabean oil, ethanol and water

Methodology : Pour a few drops of soybean oil into a test tube. Mix it with a little ethanol. Now shake the test tube very well. Then pour a little water further into the test tube and shake it again. Observe what happens. The oil solution will be of somewhat muddy colour.

Do the same experiment using mustard, coconut and sesame oil and describe what happens.

Calorie and Work Energy of Food

The three nutrients of food-carbohydrates, proteins and fats -produce heat for the body. This heat provides energy in our body. To learn about the nutritive substance and their amount, the calorie of carbohydrates, proteins and fats must be determined. In this regard,

In our body

1 gram of carbohydrate can yield 4 kilocalorie of energy

1 gram of protein can yield 4 kilocalorie of energy

1 gram of fat can yield 9 kilocalorie of energy

the calorie value of vitamins, mineral salts and water is considered zero. In our body, metabolic activities such as digestion of food, respiration, blood circulation involved.

The energy needed to run the metabolic activities is called basal metabolism. Additionally, energy is also spent during physical labour. We obtain energy from food. Again, the heat produced from food in the body is measured in calories. 1000 calories make 1 kilocalorie. Kilocalorie is the measuring unit of heat obtained from food. The requirement of energy for the body is also determined in kilocalories.

The need of calories varies for my body, yours, your younger brother and your father. In our bodies, energy is used in two ways. For example- 1. Inside the body e.g basal metabolism and 2. Physical activity. The demand of calorie or heat energy for a person depends mainly on age, height and weight of the body. Additionally, the amount of energy a person uses each day varies according to sex, body-size, occupation and special conditions such as pregnancy.

In the following chart the intake of calories and food demand is shown.

The amount of energy used each day by children, women and men of different age groups

Age (year)	Average weight (kilogram)	Average energy (kilocalorie)	Age (year)	Average weight (kilogram)	Average energy (kilocalorie)
infants			females		
0.0-5 Month	6	115	10-12	30	1900
6 Month - 1.0 year	9	100	13-15	42	2200
			16-19	51	2100
children			20-39	54	2000
1-3	13	1300	40-49	53	1900
4-6	20	1500	50-59	52	1800
7-10	28	1800	60-69	51	1600
			70+	51	1400
males			Extra		
10-12	40	2200	demand fo		
13-15	44	2500	pregnant		
16-19	67	3000	women for		
20-39	67	2700	first 3 months		+150
40-49	70	2400	2nd 3 months		+200
50-59	68	2300	3rd 3 months		+300
60-69	65	2200	Additional		
70+	65	1900	demand of		+400
			lactating mothers		

How will we know how much energy a person requires daily? It depends on three factors. 1. Metabolism 2. physical labour and 3. Influence of food.

Our daily foods should match our needs. When selecting foods, we should keep in mind that the body can obtain the necessary amount of calories from food and it should also contain vitamins, mineral salts and other essential nutrients.

Lesson 6 : Vitamins

Through different investigations the scientists have proved that food contains a number of subtle nutrients other than carbohydrates, proteins, fats and mineral salts. In their absence body suffers from diseases like night blindness, beriberi, scurvy, etc. The term vitamin refers to an essential dietary factor which is required by the organism in small amounts and whose absence results in deficiency diseases. Thus, vitamin can be defined as potent organic compounds which must be available to the animals so that specific physiological processes (growth, replacing and repairing worn-out and damaged tissues or energy production) essential to life may proceed normally. The vitamins do not have active participation in body formation.

Types of Vitamin: Vitamins are classified into two sub-groups based on their solubility properties. For example:

1. Fat soluble vitamins, e.g.–vitamin-A, vitamin-D, vitamin-E and vitamin-K.
2. Water soluble vitamins, e.g.- vitamin B-complex group and vitamin C.

Sources of vitamins: Vitamins are found in green leaves, fresh branches, yellow and green leafy vegetables, fruits and seeds, etc.

Vitamin A

Sources: It is found largely in fish liver oil and animal fats. Vegetables rich in carotene such as red amaranth (lalshak), spinach (palongshak), vine spinach (puishak), tomatoes, carrots, beets and pumpkins, etc. are good sources of vitamin-A. Different fruits like papaya, mango, and jackfruit contain vitamin-A. It is also found in small fish like carp (mola), catfish (dhela) etc.

Functions of vitamins

To maintain good eyesight, keep skin and mucus membranes healthy, protect the body from various contagious diseases, aid digestion and stimulate appetite, maintain blood in a normal state, and support nutrition and growth.

Deficiency Related Diseases

1. Night blindness: The symptom of this disease is, one cannot see in inadequate (dim) light specially at night. The children are the main victims of this disease. If it continues for a long time, one might lose his eyesight. To prevent this disease, green vegetables and colourful fruits should be added to the food menu of the children. Vitamin-A capsule can also prevent this disease. In our country, vitamin-A capsules are given to the children during vaccination days at different vaccination centres.

2. Xerophthalmia: If there is any deficiency of vitamin 'A', the cornea of one's eye may be affected. A dry layer appears on cornea. The eyes get dry and rolling of water from eyes also gets stopped. The eyes cannot tolerate light, pus accumulates in the eyes and the eye-lids become swollen. If proper treatment is done, the disease may be cured. However, if treatment is delayed, children could become blind. In addition, a vitamin A deficiency can hinder normal growth. Diseases such as cough, flu, and influenza may also occur due to this deficiency.

Vitamin B-Complex The function of vitamin B-complex group is to break down proteins, carbohydrates and fats as part of specific enzymes and help to release the energy from within.

Vitamin B1 (Thiamin): The main function of vitamin B1 is to participate in carbohydrate metabolism and release energy. Besides, it helps to maintain normal appetite and keeps the nervous system active.

Vitamin B2 (Riboflavin): It takes part in the metabolism of amino acid, fatty acid and carbohydrates and helps to produce energy.

Vitamin B6 (Pyridoxine): It helps in energy production.

Lesson 7 : Vitamin 'C'

Vitamin-C is a very essential nutrient for our body. This vitamin is water soluble and can be wasted with a little stroke of heat. Since it is not stored in the body, we should eat it everyday. It is found extensively in all citrus fruits like lemon, orange, shaddock, pine apple, guava, amlaki, green vegetables like cabbage, cauliflower, tomato, and lettuce. Fresh vegetables and fruits contain more vitamin 'C' than the ripe ones.

Vitamin 'C' strengthens muscles and teeth, heals wounds and resists skin diseases, and prevents infection in throat and nose.

Deficiency Related Diseases

In adults, an acute shortage of vitamin C shows the following symptoms.

- Structure of bones cannot be strong.
- Bones become weak and fragile.
- Skin becomes dry, itches and the healing of wound is delayed.

Scurvy

- The gums swells up and becomes soft.
- The root of the teeth become loose and eventually bleeds.
- The enamel weakens, leading to tooth loss. Children and the elderly person suffer most from this disease.
- The glands swell up and mouthache is felt.
- Bleeding does not stop easily, and healing of wounds delayed.
- Other diseases like flu, and cough easily attack the body.

Remedy

At this stage consulting a doctor is very essential.

Prevention

A baby should be given, along with breast milk, other supplementary foods like fruit juice, vegetable soup etc.

Vitamin 'D'

Vitamin 'D' is found abundantly in edible oil, milk and milk products, different fish liver oil, egg yolk, butter, ghee, fats and hilsha fish.

Functions

- Forms structure of bones and teeth.
- Increases absorption of calcium in intestine.
- Controls the extent of calcium and phosphorous during blood circulation.

Deficiency Related Diseases

The deficiency of vitamin 'D' obstructed the absorption and accumulation of iron, and hamper hemoglobin formation.

Rickets**Symptoms of rickets**

Due to the absence of vitamin 'D' and calcium the bones of the children become soft and their growth is hampered.

- The leg bones become curved like a bow and the other bones also get curved due to the pressure of the body.
- The bone joint of hands and legs swell up.
- The ribs also turn bent.

Remedy

At this stage a doctor should be consulted immediately.

Prevention

The children should be given vitamin 'D' enriched food. Vitamin D is obtained from the cholesterol of our skin under the influence of sunlight. Therefore, the children should be allowed to play in the sun for a while.

Osteomalacia

The rickets in elderly person is known as osteomalacia. The symptoms of this disease are as follows:

- Due to the deficiency of vitamin 'D' the absorption of calcium is hampered.
- The reserve of calcium and phosphorus reduces.
- There is a change in the function of thyroid gland.
- Since the bones weaken, their hardness also decreases, and as a result increasing the possibility of the bones breaking from a light stroke.

Remedy

If the above symptoms are visible, the advice of a doctor should be sought. Food containing appropriate amount of calcium and vitamin- D should be consumed. If needed, medicine should be taken for these food nutrients.

Prevention

- Ensure the consumption of vitamin -D and calcium enriched food for children.
- Children should be allowed to play in the sun for a while.

Vitamin 'E'

Edible oil is the best source of vitamin 'E'. Food grains, liver, fats of fish and meat contain Vitamin 'E'.

Functions

- Vitamin 'E' helps in cell formation.
- It takes part in some metabolic activities of the body.
- In very few cases the deficiency of Vitamin 'E' occurs and its deficiency symptoms are also negligible.

Vitamin 'K'

It is found in green vegetables, spinach, lettuce, cabbage, cauliflower, egg yolk, soyabean oil and liver.

Functions

- It produces the protein element prothrombin in the body.
- Prothrombin helps in blood coagulation.

Deficiency Related Diseases

The liver secretes bile and if there is difficulty in the secretion of bile, the absorption capacity of vitamin 'K' is reduced. The bleeding under the skin and inside the body due to the deficiency of vitamin 'K' if not stopped, may result in the death of the patient.

Due to vitamin- K deficiency bleeding of a patient, under operation is not stopped easily. Sometimes it might be fatal.

Fill in the following chequer board

Vitamin	Source	Function	Deficiency related diseases
'A'			
'B'			
'C'			
'D'			

Lesson 8 : Mineral Salt

We include salt in our daily rice and curry. Many other types of salt are also very important for our health. Like proteins, carbohydrates, and fats mineral salts do not produce energy in the body. However, salt is an essential elements for the body's cells and fluids. Food containing calcium, sodium, magnesium, phosphorus, chlorine, iodine, iron, sulphur etc. all get into the body and help in body formation. These elements are not found as basic elements in the body, they exist with other substances as organic and non-organic compounds. Mineral basically functions in two ways in our body. For example-they act as body forming elements and help control the internal activities of the body. Meat, egg, milk, green vegetables and fruits are the main sources of mineral salts.

Mineral salt forms part of the body and regulates its internal functions. For, the formation of bones, teeth, enzymes and hormones, as well as the stimulation of nerves, muscle contractions keeping the osmotic balance of the body and activating different enzymes, mineral salt is an essential element.

Necessity of Mineral Salts in Human Body

Calcium forms teeth and bones, coagulates blood and helps maintain the smooth functioning of the nervous systems. Phosphorus also forms teeth and bones and is necessary for making phospholipids. Iron forms the red blood corpuscles, helps the enzymes in their function. Iodine is essential for the thyroid gland and aids in maintaining a smooth metabolic process.

Lesson 9 : Deficiency Related Diseases

Rickets: Calcium is absorbed along with vitamin 'D' in the body. Its deficiency causes rickets. It has been described in the section of vitamin.

Goiter: It is also called ghag. The prevalence of this disease is higher in the northern part of our country especially in Rangpur, Dinajpur, Jamalpur and Mymensingh districts. In the absence of iodine in the body the thyroid gland becomes bigger and the throat gets inflated. This is called goiter or ghag.

The symptoms of this disease are as follows:

- Inflated thyroid gland.
- Sound produced during breathing.
- Hoarseness of voice.
- Feeling discomfort in the throat and trouble in swallowing of food.
- Patient feels dizziness and weakness.

Remedy

In the early stage of this disease, iodized salt, sea fish, fish liver oil and marine algae should be eaten. If necessary, a doctor should also be consulted for proper treatment.

Cretinism

Usually, children are exposed to this disease due to iodine deficiency. The symptoms of this disease are as follows:

- Growth of the body becomes slow.
- Thick skin and a deformed face.
- Thick lips, large tongue and risk of turning mentally retarded.

Remedy

Proper and timely treatment helps to overcome the problem and to maintain the normal growth of children.

Prevention

Iodized salt should be eaten regularly.

Anemia

Iron is the constituent element of haemoglobin of red blood corpuscles. People If the children and pregnant women lack iron in their food, they suffer from anemia. Usually, children attacked with round worm may experience this disease.

Symptoms

- Weakness and drowsiness of head and body.
- Unusually rapid palpitation.
- Dizziness and difficulty breathing particularly even with light activity.
- Loss of weight and appetite.

Remedy

Vegetables enriched with iron, fruits, meat, egg yolk, liver, etc. should be eaten in ample quantities. If needed a doctor may be consulted for proper medication. If the disease turns acute, the rate of blood circulation through heart may go up and one might be at the risk of heart attack following the stoppage of heart beat.

Water

Water is an essential nutrient for our livelihood. Our body contains 60- 70% water. In the formation of the body, the role of water is very much important. The water remains inside and outside the cells of bones, flesh, skin, nails, teeth etc. Almost all foods contain water. However, we meet our water needs by drinking it separately.

Apart from body formation, water controls all the internal activities of the body. Without water no chemical reaction can take place. Water acts as solvent in the body. Different mineral salts remain dissolved in water. The digestion of food goes on in presence of water. Again the food nutrients get absorbed in the body while they remain dissolved in water.

Function

- Water is responsible for blood circulation and thermal regulation of the body.
- Water releases waste materials from the body. For example urine, and sweats.

When cholera and diarrhoea occur, a great quantity of water is released through defecation and vomit. As a result, the body suffers from water deficiency. If cholera or diarrhoea occurs, the patient should be given oral saline or water-salt beverage (Shorbot). It is the easiest treatment for cholera or diarrhoea. Besides, International Center for Diarrhoeal Disease Research of Bangladesh (ICDDR) has produced packet oral saline for this disease. It is mixed with water and then given to the patient to drink. Recently a saline, named grain saline has been invented. This saline is made up of 1 litre water, 50 gm rice powder and a pinch of salt.

Task: Earlier you have learnt how to make oral saline. This time you again prepare oral saline. Write down what precautions you need to undertake while preparing oral saline.

Dryness

If there is a fall in the quantity of water in the body, the cells experience shortage of water. When the cell water is reduced one gets very thirsty, the blood pressure drops, blood circulation and digestion process are hindered. The body loses weight if there is water shortage and muscles and nerve cells become weakened. If the water level of the body goes below 20 percent, the normal body function hampered, the patient gets fainted and even may run the risk of death.

Roughage (Dietary fibre)

Roughage is the part of cereal grains, fruits and vegetables which is not digested. Roughage does not change inside the body. It is not a nutrition element. However, it is very essential to maintain a sound health. Roughage moves directly through the alimentary canal. Roughage in grains is called bran. Roughage in fruit and vegetables is the cellulose walls of plant cells. We get roughage from fibre foods.

Food Selection

The foods which meet the demand of the calorie of the body, maintain the growth and formation of tissue cells and control the metabolic activities of the body is called balanced diet. Therefore, by balanced food we mean the correct amount of food and the proper proportions of each (six) type of food that meet the demands of the body of a particular person.

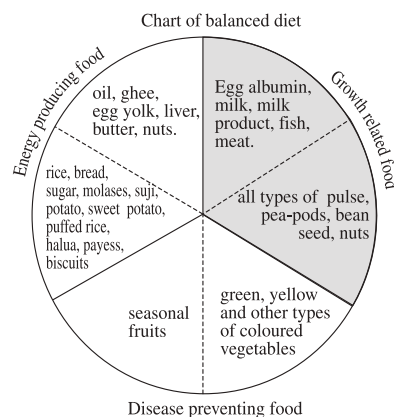
The necessary nutritional elements in food remain in specific quantities in a balanced diet depending on age, sex, physical condition, work load etc. A balanced diet chart should be made following some instructions. Such as-

1. In every single meal of the day the inclusion of six food elements including protein, carbohydrates and fats should be confirmed.
2. Food from each group has to be supplied based on age, sex and occupation.
3. Out of the total daily intake of calorie, 60-70% should come from carbohydrates, 10% from proteins and 30-40% from fats.

List of Balanced Diet

Following some guidelines we can make the list of balanced diet. For example:

1. Firstly the list has to be made based on the understanding that the requirements of different food elements vary according to age, occupation, and physical status of a particular person.
2. Meeting the calorific value of food according to physical needs of one's body.
3. The food should contain correct proteins needed for body formation and the prevention of its decay.
4. Ensure the presence of appropriate vitamins, mineral salts and water in food.
5. To acquire knowledge about the nutritional value and classification of food. First we have to select food from the main divisions of food. There should be variety in food selection.
6. There should be awareness about food habit while selecting food.
7. The food list has to be prepared matching with the financial ability of an individual and a family as well.
8. While preparing the list, season and weather condition should be taken into consideration.



New words : Protein, carbohydrate, lipid, vitamin, easily digestible quotient, amino acid, kwashiorkor, merasmus, xerophthalmia, scurvy, rickets, osteomalacia, prothrombin, cretinism, anemia.

What we have learned at the end of this chapter-

- The fundamental metabolism is the energy required to run metabolic process.
- Vitamins and mineral salts are not separate types of food. These all are obtained from other food-stuffs.
- Water is a very essential nutrient for the body. The digested foods, being dissolved in water, is transported to all parts of the body.

Sample Questions

Multiple Choice Questions

1. Which one produces heat and energy of the body?
 - a. Water
 - b. Vitamin
 - c. Fats
 - d. Mineral salts
2. The deficiency of which vitamin causes rickets in children?
 - a. Vitamin 'A'
 - b. Vitamin 'C'
 - c. Vitamin 'D'
 - d. Vitamin 'E'

Read the following paragraph and answer questions no. 3 & 4

Sumi does not like sour. Even she does not like green vegetables and tomatoes. Recently she has been experiencing bleeding from her gums.

3. What disease has attacked sumi?
 - a. Scurvy
 - b. Rickets
 - c. Merasmus
 - d. Kwashiorkor?
4. What happens to the elderly person due to the deficiency of foods mentioned in the paragraph?
 - i. the bones becomes soft
 - ii. the skin itches and wounds
 - iii. pain in thoracic cage and ribs appears

Which one is correct?

- a. i & ii
- b. i & iii
- c. ii & iii
- d. i, ii & iii

Creative Questions

1. Sara nowadays doesn't want to eat anything. She has lost her appetite and often feels nauseous. Her skin is becoming rough. When she consulted a doctor, she was advised to eat more eggs and milk. Meanwhile, her grandmother is suffering from bone weakness due to a calcium deficiency.

- a. What is food?
- b. Why is Nyctalopia or night blindness dangerous? Explain.
- c. What disease is Sara's grandmother suffering from? Explain.
- d. If Sara does not follow the doctor's advice, what problems might she face in the future? Analyze.

2. Nurjahan Begum is very concerned about the physical growth of her 8-year-old son Bakul because he refuses to eat meat, milk, eggs, or fruits. As a result, he lacks the nutrient needed in various biological processes in his body. To ensure his growth and good health, she started giving him special types of food. However, she chose different foods for herself and for Bokul's father, grandfather, and grandmother.

- a. What is protein?
- b. Though roughage is not a nutrient, how it helps maintain health. Explain.
- c. Explain the function of the nutrient that Bakul is lacking.
- d. Analyze the logic behind Nurjahan Begum's different food choices for family members.

Short Answer Questions

1. Why is it good to expose children to sunlight for a short time? Explain.
2. Why is goiter more common in northern areas? Explain.
3. What is the function of Vitamin K? Explain.
4. What can happen due to iron deficiency in the blood? Explain.